

I Don't Believe Misinformation, But I Might Share It: Inoculation Effects in Messenger Communication

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Abstract

Mobile instant messaging services (MIMS) such as WhatsApp and Telegram are used to spread misinformation, yet evidence on effective countermeasures within these spaces is limited. Guided by inoculation theory, we conducted an online experiment ($N=154$) to examine the extent to which refutational-same (argument-matched) inoculation or cross-protection (cross-topic) conditions influence the credibility and sharing of false and true information within MIMS. We analyzed whether critical thinking abilities, information literacy, and messenger use moderate the effectiveness of these measures. Results show that refutational-same inoculation only reduced the perceived credibility of misinformation but did not lower sharing intentions, whereas no cross-protection effects were observed. Overall, inoculation had no effect on credibility or sharing of true information. No moderators were detected across critical thinking measures, information literacy, or messenger use, although political orientation consistently predicted perceived credibility and sharing intentions of misinformation. These results suggest that inoculation messages within MIMS might be more effective when they are content-specific.

Keywords

misinformation, messenger communication, inoculation, credibility, sharing intention, critical thinking, information literacy

Introduction

Misinformation circulates widely online and can shape beliefs and behavior with societal costs (e.g., Bovet & Makse, 2019; Kouzy et al., 2020; Vosoughi et al., 2018). Increasingly, this circulation occurs not only on public social media feeds but also within mobile instant messaging services (MIMS) such as WhatsApp and Telegram, which now serve for news sharing alongside interpersonal exchange (Goh et al., 2019). MIMS enable rapid, informal sharing and out-of-context content can mislead and accelerate misinformation (Chen et al., 2015; Qian et al., 2022; Sundar et al., 2021). Moreover, end-to-end encryption protects users but limits oversight (De Freitas Melo et al., 2020; Swart et al., 2018). These affordances can enable counter-publics to circulate their own narratives, including misinformation, eroding trust and fuelling polarization (Panahi et al., 2025). Thus, effective interventions against misinformation within MIMS are urgently needed. Consistent with prior work, we define *misinformation* as false or misleading information that is shared regardless of intent (Altay et al., 2023).

One promising approach is psychological inoculation, which aims to build resistance to persuasion by pre-exposing individuals to weakened counterarguments and refutations (McGuire, 1981/1964). Originally developed to protect attitudes against

persuasive attacks, inoculation has been applied across domains and generally promotes resistance to subsequent influence attempts, including those involving misinformation (Banas & Rains, 2010; Lu et al., 2023). Refutational-same inoculation pairs a weakened version of a specific attack argument with an immediate refutation, such that the same arguments are later encountered in the persuasive message (McGuire, 1981/1964). A related line of research examines cross-protection, referring to cases in which resistance induced for one issue generalizes to related but unaddressed topics (Parker et al., 2012). Whether these message strategies also reduce susceptibility to persuasive attacks in the form of misinformation within MIMS, and how specific they must be to be effective, remains an open question.

The present study addresses this gap by testing refutational-same inoculation and cross-protection effects in a simulated group chat using a mixed design with three groups: two experimental conditions (each exposed to

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refutational-same and cross-protection conditions) and one control (no inoculation). We examine whether these inoculation messages reduce the perceived credibility and sharing intentions associated with subsequent messages, including false and true content. Importantly, disbelieving true information is also a growing concern; effective interventions should not merely increase general skepticism. We also assess potential boundary conditions by testing whether effects vary with individual differences such as critical thinking abilities, information literacy, and reliance on messenger services as a primary news source.

Our study seeks to clarify whether and how inoculation-based message structures operate in private messenger environments. The results inform under which conditions such approaches may reduce susceptibility to persuasive misinformation and whether they can ultimately curb the spread of misinformation within MIMS.

Theoretical Background

Misinformation in Messenger Communication

MIMS enable the exchange of text, voice messages, images, videos, and more in group or one-to-one conversations. MIMS are increasingly used to share news with friends or acquaintances (Goh et al., 2019), which can lead to news being circulated via screenshots or stripped of context (Chen et al., 2015; Sundar et al., 2021). In contrast to other social media platforms (e.g., Instagram), the original source of information is often not visible because of technical constraints (Rossini et al., 2021). When circulating outside its original context, shared content can mislead and, alongside other MIMS-specific features, increase the dissemination of misinformation (Hameleers et al., 2020; Qian et al., 2022). MIMS also differ in terms of privacy and transparency: public platforms are perceived as more open and observable, whereas MIMS are seen as private and intimate (Malhotra, 2020). This stems in part from end-to-end encryption offered by many MIMS (e.g., WhatsApp), which makes it difficult to combat the influence of misinformation (De Freitas Melo et al., 2020) and limits external oversight of what is shared and how exchanges unfold (Swart et al., 2018). Previous measures have proven ineffective: limiting message sharing can slow dissemination but offers limited benefit when information can be shared within groups of hundreds (De Freitas Melo et al., 2020). Therefore, different approaches against misinformation in MIMS are urgently needed.

Much research on combating misinformation focuses on post-exposure corrections like *debunking* (e.g., Chan et al., 2017). However, misinformation can continue to influence judgments even after it has been corrected, consistent with the continued influence effect (Gordon et al., 2017; Johnson & Seifert, 1994). Psychological inoculation offers a complementary approach by strengthening resistance before exposure, thereby reducing susceptibility to later misleading messages.

This may be especially suitable for MIMS, where encrypted and fragmented sharing makes it difficult to trace exposure and reliably reach recipients with timely corrections. Moreover, when information is shared through interpersonal networks and outside of its original context, building resistance prior to exposure may be especially valuable.

Psychological Inoculation

McGuire's inoculation theory proposes that attitudes can be protected against persuasion much like health can be protected by vaccines (McGuire, 1981/1964). Inoculation theory was originally formulated as a general theory of resistance to persuasive attack and is content-agnostic. Early work focused on defending truisms (Compton, 2013; McGuire, 1981/1964), while later research applied inoculation across domains such as health, politics, conspiracy beliefs, and also misinformation (Banas & Miller, 2013; Lu et al., 2023; Pfau & Burgoon, 1988; Pfau et al., 1992). The theory's biomedical analogy is central: recipients encounter a weakened version of an attack argument that is immediately refuted by counterarguments (Compton, 2013; McGuire, 1981/1964). This process is intended to stimulate cognitive "mental antibodies" (Lewandowsky & Van Der Linden, 2021, p. 10) without changing underlying attitudes. Messages typically include (a) a forewarning of an impending attack and (b) a refutational preemption (*prebunking*), which previews and counters the anticipated attack (Compton & Pfau, 2005). This combination is expected to heighten perceived threat and prompt counterarguing, shifting processing from heuristic to more reflective evaluation (Cook et al., 2017). Although refutational preemption represents a common and well-established inoculation format, alternative approaches have also relied on credibility-based, affective, or metacognitive strategies (e.g., Anderson, 1967; Clyne et al., 2020; Pfau et al., 2006).

Evidence shows that inoculation fosters resistance to persuasive attacks (Banas & Rains, 2010). Importantly, these "attacks" need not involve misinformation but may consist of any counter-attitudinal argument. The theoretical mechanism itself is not restricted to false content, although such attacks may take the form of misinformation. Applied to the misinformation context, recipients are expected to exhibit greater resistance to persuasive misinformation, which may be reflected in lower credibility judgments and reduced sharing intentions.

Inoculation messages that follow the structure above are also referred to as *narrow-spectrum* inoculations, in which specific counterarguments are previewed and refuted prior to exposure to a later persuasive attack (Lewandowsky & Van Der Linden, 2021). The term "narrow-spectrum" refers to the design of the inoculation message (i.e., argument-specific defense), rather than the scope of its eventual protective effects. Within the classical inoculation framework, the distinction between refutational-same and refutational-different is defined by the correspondence between the arguments

raised in the inoculation message and those presented in the subsequent attack (McGuire, 1981/1964). In refutational-same conditions, the later attack relies on the same arguments that were previewed and refuted in the inoculation message. In refutational-different conditions, the later attack introduces novel arguments against the same attitude that were not explicitly addressed in the initial defense. A related but conceptually distinct line of research examines cross-protection, referring to cases in which resistance induced for one issue generalizes to related but unaddressed issues (Parker et al., 2012, 2016). Compton and Pfau (2005) describe this phenomenon as “umbrella protection,” whereby strengthening defenses for one attitude may spill over to related attitudes. Empirically, inoculating on one topic (e.g., contraception) can reduce openness to persuasion on a related domain (e.g., binge drinking), and inoculating on one health issue can generalize to others (Parker et al., 2012, 2016). The prevailing interpretation is that refutational preemption offers guided practice in counterarguing (i.e., teaching how to counter, not just what to counter), thereby enabling broader resistance beyond the specific content addressed (Parker et al., 2016). At the same time, the mechanisms and boundary conditions of such cross-protection remain under-specified, warranting tests with weakly related topics and improved measurement (Compton et al., 2021; Parker et al., 2016).

A related approach, *broad-spectrum* inoculation, teaches general manipulation techniques so that varied future attacks are recognized (Maertens et al., 2021; Roozenbeek et al., 2022b). Examples of common techniques include emotional language and scapegoating (Roozenbeek et al., 2022b) or relying on fake experts to boost the credibility of misinformation (Cook, 2020). It remains debated whether such approaches reliably elicit perceived threat and motivational resistance processes (i.e., core features of classical inoculation) or whether they function primarily by enhancing critical thinking skills. Approaches that incorporate threat-inducing forewarnings against upcoming misinformation attacks may therefore still align with core inoculation mechanisms, even when they emphasize general manipulation techniques (Roozenbeek et al., 2022b).

The present study focuses on *narrow-spectrum* inoculation and examines both argument-matched effects (i.e., refutational-same) and potential cross-protection generalization of resistance within a messenger communication context.

Psychological Mechanism. Inoculation is posited to work mainly through forewarning and refutational preemption, which activates a sense of threat and counterarguing (Compton, 2013; McGuire, 1981/1964). Evidence generally supports a role for perceived threat, although meta-analytic results are mixed (Banas & Rains, 2010; Ivanov, 2017; see also Compton & Pfau, 2005). Refutational preemption then provides a template for counterarguing: recipients practice how to counterargue likely attacks and can later generate their

own counterarguments, individually or in conversation (Compton & Pfau, 2005; Ivanov, 2017). Beyond structured counterarguing, inoculation may also operate through affective responses (e.g., anger), increased attitude accessibility, or credibility-related processes, indicating that its effects are not limited to purely logical rebuttal (Compton & Pfau, 2005). Secondary contributors, such as issue involvement and elaboration, often increase following inoculation and may bolster resistance by sustaining issue-relevant thinking, although effects tend to be modest and sometimes inconsistent (e.g., Compton et al., 2021). In summary, inoculation seems to signal vulnerability and trains defense. Whether and how these processes operate within MIMS remains unclear.

Effectiveness of Inoculation

The present study examines how inoculation-based messages influence credibility assessments and sharing intentions for both false and true information within MIMS. In addition to testing the effects of refutational-same inoculation and cross-protection conditions, we examine whether these effects vary as a function of individual differences such as critical thinking abilities, information literacy, and reliance on MIMS as a primary news source.

Misinformation: Credibility and Sharing. Inoculation confers resistance across multiple persuasive contexts, including misinformation (Banas & Miller, 2013; Jolley & Douglas, 2017; Lu et al., 2023). Empirically, detailed refutational inoculation reduced susceptibility to climate-change misinformation (Van Der Linden et al., 2017) and buffered opinion change in response to astroturfing comments when the inoculation mirrored the later attack (Zerback et al., 2021). A meta-analysis across 42 studies concludes that inoculation reliably increases resistance to misinformation, although many included interventions were technique based (Lu et al., 2023).

Most existing studies tested inoculation in social media like settings. Little is known about effectiveness within messenger services, where content is private, rapid, and often out of context, and where the original source of information is frequently unclear or difficult to verify. Accordingly, we focus on the perceived credibility of the message content rather than on source credibility. Building on the general pattern that inoculation reduces perceived credibility of misinformation, we expect a similar effect in messenger communication:

Hypothesis 1 (H1): Participants who receive a refutational-same inoculation before encountering misinformation rate the misinformation as less credible than participants without inoculation.

A practical constraint is that exact message matching is rare. Accordingly, it is important to test cross-protection effects. Evidence is mixed: topic-matched formats often yield stronger protection (Zerback et al., 2021), whereas

there are generalized but sometimes smaller effects when inoculation and attack do not match (Roozenbeek et al., 2022a). Guided practice in rebuttal (via refutational preemption) may nonetheless transfer to novel arguments by motivating defense and scaffolding counterarguing (Compton, 2013; Compton et al., 2021).

Hypothesis 2 (H2): Participants who receive an inoculation on a different topic (i.e., cross-protection condition) before encountering misinformation rate the misinformation as less credible than participants without inoculation.

Beyond lowering perceived credibility, curbing the spread of false messages is crucial. Platform level measures on messenger services show limited impact (De Freitas Melo et al., 2020). Inoculation may help in reducing sharing by decreasing perceived credibility (Halpern et al., 2019) and by prompting more reflective processing that encourages pausing before forwarding (Cook et al., 2017). However, credibility is not the only driver of sharing, since altruistic motives and social factors can promote diffusion even when accuracy is doubted (Apuke & Omar, 2020). Piltch-Loeb et al. (2022) found that narrow-spectrum, broad-spectrum, and combined inoculation reduced intentions to share COVID-19 misinformation. A meta-analysis reported reductions for health-related misinformation but had no reliable effect on general sharing intentions across topics (Lu et al., 2023). Given sparse and mixed results, we ask:

Research Question 1 (RQ1): How does a refutational-same inoculation delivered before exposure affect people's intention to share misinformation, compared with no inoculation?

True Information: Credibility and Sharing. Conceptually, inoculation should not lower credibility judgments for true content because reflective processing helps recipients recognize valid claims (Cook et al., 2017) – although evidence is mixed. Some studies find no decrease in belief for true information (Roozenbeek & Van Der Linden, 2019), whereas meta-analyses diverge: Lu et al. (2023) find no overall reduction in belief for true information, although Modirrousta-Galian and Higham (2023) report that gamified broad-spectrum inoculations do not improve veracity discernment and may nudge judgments toward caution. In private, context poor messenger settings, both outcomes are plausible. Reflective processing may preserve credibility for true information, yet heightened caution may slightly depress assessments. We therefore ask:

Research Question 2 (RQ2): Does presenting a refutational-same inoculation prior to exposure affect the

perceived credibility of a true message, relative to no inoculation?

Whether inoculation affects sharing intention of true information is less examined. Evidence from technique based, broad-spectrum interventions generally show no reduction (Basol et al., 2021). In a related video-based study, participants shared less false and more true information (Roozenbeek et al., 2022b). Meta analytic results likewise show no harm: inoculation does not decrease intentions to share real information (Lu et al., 2023). Thus, refutational-same inoculation should also not dampen sharing intention of true messages within MIMS.

Hypothesis 3 (H3): Participants who receive a refutational-same inoculation before a true message do not report lower intention to share the true message compared with participants without inoculation.

Potential Moderators of Refutational-Same Inoculation Effects

Several individual differences and use patterns may shape refutational-same inoculation effectiveness. Because inoculation is thought to prompt reflective, effortful processing through forewarning and refutational preemption, recipients with stronger dispositions or skills for such processing may benefit more, whereas habitual reliance on messenger services for news may attenuate or alter the effect. We therefore briefly introduce potential moderators and the corresponding hypotheses.

Critical Thinking. Critical thinking can be defined as “the propensity and skill to engage in an activity with reflective skepticism” (McPeck, 1981, p. 8). To the extent that inoculation activates reflective or analytical processing, its impact may partly depend on individuals’ baseline propensity and skills for such thinking. These skills include cognitive reflection, the disposition to think critically, and the motivation to do so (i.e., need for cognition).

Cognitive Reflection and Critical Thinking Disposition. Performance on the Cognitive Reflection Test (CRT; Frederick, 2005) is negatively associated with the perceived credibility of misleading headlines and positively with headline discernment (Pennycook & Rand, 2019, 2020). Critical thinking disposition predicts better detection of misinformation on social media (Orhan, 2023) and its recognition (Escolà-Gascón et al., 2021). Even brief stop-and-think prompts can reduce trust in false claims (Kruijt et al., 2022). Individuals with higher cognitive reflection and stronger disposition toward critical thinking may be more likely to engage with and apply the counterarguments provided in inoculation messages when evaluating subsequent misinformation. Accordingly:

Hypothesis 4 (H4): The relationship between presenting a refutational-same inoculation and the perceived credibility of misinformation is moderated by cognitive reflection and critical thinking disposition (i.e., stronger credibility reductions at higher reflection and higher disposition).

Need for Cognition (NFC). NFC is the tendency to engage in and enjoy effortful thinking (Cacioppo & Petty, 1982). Higher NFC is linked to greater cognitive effort, less reliance on heuristics, and more information seeking and consideration of relevant evidence (Cacioppo et al., 1996; Metzger et al., 2015; Mokhtari et al., 2013; Petty et al., 2009). Empirically, it is generally associated with lower belief in misinformation and better accuracy judgments (Leding & Antonio, 2019; Schaewitz et al., 2020; Wu et al., 2023), with some exceptions under motivated reasoning (Borah, 2022). If inoculation prompts deep processing of refutations, those higher in NFC should better engage with and apply refutation logic to subsequent misinformation.

Hypothesis 5 (H5): The relationship between presenting a refutational-same inoculation and the perceived credibility of misinformation is moderated by NFC (i.e., stronger credibility reductions at higher NFC).

Information Literacy (IL). Complementing critical thinking, IL concerns the capacity to find, evaluate, and use information and sources (ACRL; Jones-Jang et al., 2021). While related to media literacy, IL emphasizes acting on information rather than expression through media (Livingstone et al., 2014). Historically rooted in librarianship, IL now centrally involves judging source credibility and evidence quality (De Paor & Heravi, 2020). Conceptually, stronger IL should reduce susceptibility to falsehoods by promoting careful scrutiny of claims and sources (e.g., Khan & Idris, 2019). Empirically, higher IL predicts better identification of misinformation (Jones-Jang et al., 2021). Recent work has also linked inoculation theory to metacognitive learning, suggesting that inoculation may support deeper learning processes across cognitive, affective, behavioral, and metacognitive domains (Compton, 2019). If inoculation invites reflective evaluation via forewarning and refutational preemption, individuals high in IL should be better able to apply the counterarguments provided in the inoculation message.

Hypothesis 6 (H6): The relationship between presenting a refutational-same inoculation and the perceived credibility of misinformation is moderated by IL (i.e., stronger credibility reductions at higher IL).

Messenger Services as Primary News Source. Messenger usage patterns may also shape susceptibility to misinformation. Heavy social media use, especially as a news source, correlates with belief in conspiracy theories and hoaxes

(Allington et al., 2021; Enders et al., 2023; Su, 2021). Likely mechanisms include frequent exposure to unvetted content, limited fact checking, and fatigue or overload that encourages low effort sharing (Bermes, 2021; Olan et al., 2022; Talwar et al., 2019). Messenger services intensify these risks: misinformation circulates widely with few verification affordances and may be reinforced (Bovet & Grindrod, 2022; Cinelli et al., 2021; Galhardi et al., 2020). If inoculation depends on noticing and reflectively processing forewarnings and refutations, primary reliance on messenger apps for news may blunt its benefits through habituation to rapid, context poor content and higher cognitive load.

Hypothesis 7 (H7): The relationship between presenting a refutational-same inoculation and the perceived credibility of misinformation is moderated by using messenger services as the primary news source (i.e., weaker credibility reductions among messenger-primary users).

Political Orientation

Prior research links political orientation to belief in misinformation. For instance, conservative voters are more likely to accept misinformation (Baptista et al., 2021) and are less accurate at distinguishing true from false health-related information (Calvillo et al., 2020). People's (political) preferences can influence information processing, a phenomenon also known as motivated reasoning (Kunda, 1990; for an overview, see Epley & Gilovich, 2016). Regarding misinformation, political preferences shape accuracy assessments (Tsang, 2021) and politically aligned misinformation is more likely to be seen as trustworthy (Thaler, 2024).

Political orientation has also been examined in inoculation research, where party identification and ideological alignment have been found to moderate resistance to political attack messages and the effectiveness of inoculation treatments (e.g., An & Pfau, 2004; Brinson, 2022; Pfau & Burgoon, 1988). Because the present study does not derive ideology-specific hypotheses but focuses on misinformation resistance more broadly, political orientation is included as a covariate to account for potential differences in credibility assessments.

Method

All hypotheses, the experimental design, and measures used were preregistered.¹ Data and material are available on the Open Science Framework (OSF).²

Experimental Design

We employed a mixed three-group design. In the two experimental groups, participants saw both refutational-same and -different inoculations, and in the control group, they saw no inoculation. The dependent variables were credibility assessment and sharing intention. Critical thinking abilities,

Table 1. Presented Information and Inoculation Message by Group.

Group	Veracity	Topic	Inoculation message
Group 1 (experimental)	False	Climate Change	Refutational-same
		Ukrainian Refugees	Cross-protection
	True	COVID-19 Vaccination	Refutational-same
		Microplastics	Cross-protection
Group 2 (experimental)	False	Climate Change	Cross-protection
		Ukrainian Refugees	Refutational-same
	True	COVID-19 Vaccination	Cross-protection
		Microplastics	Refutational-same
Group 3 (control)	False	Climate Change	No inoculation
		Ukrainian Refugees	No inoculation
	True	COVID-19 Vaccination	No inoculation
		Microplastics	No inoculation

information literacy, and messenger use were analyzed as moderators. Political orientation was analyzed as control variable. In every group, screenshots of a WhatsApp-like group chat conversation were presented (see Supplemental Material, S1–S3 Mockup Messenger Chat).

Each experimental condition contained two inoculation messages embedded in the chat, one paired with a true message and one paired with a false one. The inoculation messages were always presented before the thematically corresponding target information. Experimental Group 1 received inoculations for COVID-19 vaccination (true) and climate change in Sweden (false). Experimental Group 2 received inoculations for microplastics (true) and financial benefits for Ukrainian refugees (false). The control group received the same chat without any inoculations. Because each experimental group only received two inoculations, the remaining (uninoculated) items in that group functioned as cross-protection exposure. Additional true, filler messages (neutral information) were included in all chats to reduce salience of the inoculation and target items. Table 1 summarizes the presented information, its veracity, and the inoculation exposure by group.

Procedure

In the beginning of the study participants completed moderator measures in the following order: Cognitive Reflection Test (CRT), Critical Thinking Disposition Scale (CTDS), Need for Cognition (NFC), Information Literacy (IL), use of messenger services for news, and Political Orientation Scale (POS). Participants were then informed that they were members of a messenger group that periodically shared news and socially relevant messages. They viewed a simulated messenger chat (screenshot-based). The control group saw the chat containing the two true and two messages without inoculations; the experimental groups saw the same chat with two additional inoculation messages inserted before their corresponding target items (see Table 1). A manipulation check and two attention checks followed the chat exposure.

Participants then rated the credibility of each true and false item and their intention to share each item. Finally, they reported age, gender, and educational attainment. At the end, participants were fully debriefed, informed about the study's true purpose, and corrections to the misinformation.

Stimulus Material

Participants viewed a screenshot of a simulated group chat modeled after common messenger interfaces (e.g., WhatsApp). The chat included true, false, and distractor messages; depending on condition, it also included inoculation messages. Importantly, the inoculation messages (including the refutational preemption) contained factually accurate information throughout; the distinction between “true” and “false” refers to the subsequent target (“persuasive attack”) messages presented in the chat. Each chat message showed the sender's name and a timestamp and followed a natural conversation flow. German originals of all screenshots are presented in the Supplemental Material (S1 Stimulus Material).

Inoculation Messages. The inoculation messages were operationalized in accordance with the two-component structure of inoculation theory, incorporating both an explicit forewarning (to signal potential misinformation) and a refutational preemption (to provide counterarguments), as described in prior inoculation research (e.g., Compton, 2013; McGuire, 1981/1964). This combination has been shown to elicit perceived threat in prior inoculation research (Compton & Ivanov, 2012). Participants in the two experimental conditions received topically matched inoculation messages before the corresponding target information (true or false).

Forewarning (Identical Across Topics). “Dear users, various messages are shared on messenger services, often including misleading and persuasive messages that can influence you.”

The forewarning was intentionally formulated in a general manner to reflect the informal nature of messenger

communication and to avoid anchoring attention to a single topic, as multiple issues were presented within the same chat. Topic-specific threat cues were embedded within the subsequent refutational preemption.

Refutational Preemption (Topic-Specific). The refutational preemption (a) briefly restated the claim pattern likely to be encountered and (b) refuted it with arguments/facts.

Example (true; COVID-19 Vaccination):

A lot of false news about COVID-19 vaccination has been circulating for some time. However, many of the reports are false. Many news reports cast doubt on the effectiveness of the vaccination and criticize its side effects. However, it has been shown that serious side effects are very rare and only occur in 0.02 percent of all cases.

Example (false; Ukrainian Refugees): “There is currently a lot of false news circulating about financial and social benefits for refugees in Germany. Much of the news is false, with war refugees, local people and recognized asylum seekers all receiving the same state benefits.”

Complete translated inoculation messages can be found in the Supplemental Material (S4 Presented Inoculation Messages).

True and False Information. Each participant was exposed to four target messages: two true and two false. True messages contained true information about microplastics and the COVID-19 vaccination. False messages presented false information about climate change and Ukrainian refugees.

For example (true; COVID-19 vaccination): “Hello everyone, I recently heard something interesting about the Covid-19 vaccination. The results of a retrospective study involving over a million children show that a Covid-19 vaccination can also protect children and adolescents against long Covid.”

For example (false; Ukrainian refugees): “I can’t believe what I’ve just read. Ukrainian refugees are to receive a much longer period of eligibility and higher BAföG³ amounts than other students. According to this, Ukrainian refugees receive almost twice as much BAföG as everyone else.”

Distractor messages unrelated to the four topics were presented to reduce the salience of inoculations and targets and to maintain a natural chat rhythm. Complete translated true and false messages can be found in Supplemental Material (S5 Presented True and False Messages).

Measures

Moderator Variables

Critical Thinking. Critical thinking was assessed with three measures – one targeting objective ability (CRT), one targeting subjective disposition (CTDS), and one targeting subjective motivation (NFC).

CRT. The multiple-choice CRT (Sirota & Juanchich, 2018) comprised the three original items (Frederick, 2005) plus four additional items (Toplak et al., 2014). An example question includes “A bat and a ball cost 1.10€ in total. The bat costs 1.00€ more than the ball. How much does the ball cost? [Correct answer = 5 cents; intuitive answer = 10 cents].” Each correct response was scored 1 (range = 0–7; $M = 4.58$, $SD = 1.96$). Internal consistency was questionable ($\alpha = .631$).

CTDS. The 11-item CTDS (Sosu, 2013) was rated on a 5-point Likert-type scale (1 = *strongly disagree* to 5 = *strongly agree*). An example item includes “I usually check the credibility of the source of information before making judgments.” The original two-factor structure (*Critical Openness*; *Reflective Skepticism*) was not supported by factor analysis; therefore, we computed a unidimensional total by summing the 11 items (range = 11–55; $M = 44.83$, $SD = 4.38$). Higher scores indicate greater critical thinking disposition. Internal consistency was acceptable ($\alpha = .729$).

NFC. NFC was measured with a six-item short scale (NFC-6; Lins De Holanda Coelho et al., 2020; adapted from Cacioppo et al., 1984) on a five-point Likert-type scale (1 = *extremely uncharacteristic of me* to 5 = *extremely characteristic of me*; $M = 3.42$, $SD = 0.69$). An example item (R) includes “Thinking is not my idea of fun.” Scores were averaged; higher values indicate higher NFC. Internal consistency was good ($\alpha = .817$).

IL. IL was assessed with 17 items (Serap Kurbanoglu et al., 2006) on a seven-point Likert-type scale (1 = *almost never true* to 7 = *almost always true*; $M = 5.22$, $SD = 0.91$). An example item includes “Critique the quality of my information seeking process and its products.” Scores were averaged; higher values indicate better IL. One instructed-response item was embedded as attention check. Internal consistency was excellent ($\alpha = .922$).

Use of Messenger Services. Frequency of using messenger services as a news source was measured with a single item adapted from Allington et al. (2021): “How often do you use messenger services (e.g., WhatsApp or Signal) to get current information or news?” Participants answered on a six-point scale (1 = *I don’t use messenger services* to 6 = *once an hour or more*; $M = 4.16$, $SD = 1.28$). (single item; internal consistency not applicable.)

Dependent Variables

Message Credibility. Perceived credibility of each true and false message was measured with a three-item semantic differential (Vraga et al., 2019; *credible – not credible*, *accurate – inaccurate*, and *informative – uninformative*) with five response levels. Items were averaged for true ($M = 3.70$, $SD = 0.71$) and false information ($M = 2.33$, $SD = 0.85$) separately; higher scores

indicate higher credibility. Internal consistency was acceptable (*true*: $\alpha = .725$; *false*: $\alpha = .681$).

Sharing Intention. Intention to share each true and false message was measured with a self-constructed three-item scale on seven-point Likert-type responses. Two items ranged from 1 = *very unlikely* to 7 = *very likely*; the third from 1 = *very small/not at all* to 7 = *very large*. Items were averaged for true ($M = 2.15$, $SD = 1.26$) and false ($M = 1.58$, $SD = 1.03$) information separately; higher scores indicate higher sharing intention. For example, “*How likely would you be to pass on the information you have just received to one or more people close to you (e.g., friends or family)?*” Internal consistency was good (*true*: $\alpha = .854$; *false*: $\alpha = .860$).

Control Variable

Political Orientation. Political orientation was measured with 28 items (POS; Decker et al., 2023) on a five-point Likert-type scale (1 = *totally disagree* to 5 = *totally agree*), comprising five factors: *sustainability* (six items; $M = 3.81$, $SD = 0.90$, $\alpha = .912$), *economics & personal security* (eight items; $M = 4.10$, $SD = 0.57$, $\alpha = .835$), *tradition* (six items; $M = 2.74$, $SD = 0.85$, $\alpha = .856$), *anti-immigration* (four items; $M = 2.67$, $SD = 1.18$, $\alpha = .921$), and *equality* (four items; $M = 4.18$, $SD = 0.67$, $\alpha = .768$). An example item includes “*I am worried about climate change.*” One instructed-response item was embedded as an attention check. Factor analysis revealed a two-factor structure: Factor 1 loaded on economics & personal security, tradition, and anti-immigration (interpreted as *conservative*); Factor 2 loaded on equality and, more modestly, sustainability (interpreted as *progressive*). Accordingly, we created two higher-order composites by averaging the relevant subscales: *Conservatism* ($M = 3.17$, $SD = 0.71$, $\alpha = .723$; higher = more conservative) and *Progressivism* ($M = 3.99$, $SD = 0.69$, $\alpha = .693$; higher = more progressive). These two composites were used in the analyses.

Manipulation Checks

Inoculation Message. We assessed whether participants noticed the inoculation by asking if they had seen any warning about potentially persuasive or misleading information in the chat. The item was dichotomous (*yes/no*). All participants answered correctly.

Message Content. Participants answered two multiple-choice questions (four options each) about the chat’s content (e.g., “*Which European country has the highest per-capita packaging-waste consumption?*”). Participants who answered incorrectly were excluded ($N = 18$).

Sample Size Rationale

We conducted an a priori power analysis in G*Power for a multivariate analysis of variance (MANOVA; global effects) with $\alpha = .05$, power $(1 - \beta) = .95$, and effect size $f^2(V) = 0.0625$.

The analysis indicated a minimum of $N = 153$ participants. Anticipating exclusions, we targeted $N = 180$ and allocated participants evenly across the three conditions.

Participants

Participants were recruited via Prolific and compensated. Eligibility required being 18 or older, proficient in German, and a WhatsApp user. We included two attention checks (before the manipulation) and two manipulation checks. Of 180 participants, 29 failed the attention checks and/or final manipulation check and were excluded, yielding a final sample of $N = 154$.

The sample comprised 49 female (31.8%), 99 male (64.3%), and 6 diverse (3.9%) participants. Mean age was 31.39 years ($SD = 9.40$; range 18–68), and all resided in Germany. Regarding education, 78 participants had a university degree (50.6%), 48 had high school diplomas (31.2%), 7 had vocational diplomas (4.5%), 15 had secondary school certificates (9.7%), 2 had no certificates (1.3%), 1 was still in school (0.6%), and 3 did not specify (1.9%).

Data Analysis

Analyses were conducted in R (Version 4.3.2). Because each experimental group contained both refutational-same and cross-protection conditions, the data were reshaped to create a three-level categorical variable for inoculation (refutational-same, cross-protection, control). To test H1–H3, RQ1, and RQ2, separate MANOVAs were conducted, using Pillai’s trace ($\alpha = .05$). To test H4–H7, moderation analyses were conducted in PROCESS for R (Version 4.3.1; Hayes, 2024).

Results

Effects of Inoculation on Misinformation

Two MANOVAs tested whether refutational-same (H1) or cross-protection (H2) conditions administered before misinformation reduced its perceived credibility and willingness to share it (RQ2), relative to a control condition, controlling for political orientation. The plots in Figure 1 show participants’ credibility assessment and sharing intention after seeing refutational-same, cross-protection, or no inoculation.

H1 and RQ1: Refutational-Same Inoculation. To test H1 and RQ1, an MANOVA was conducted comparing the refutational-same inoculation condition with the control condition on credibility and sharing intention for misinformation, controlling for progressivism and conservatism. The multivariate effect of inoculation was significant, Pillai’s trace = .030, $F(2, 200) = 3.10$, $p = .047$, $\eta^2 = .03$. Both political orientation covariates were also significant at the multivariate level, progressivism: Pillai’s trace = .061, $F(2, 200) = 6.53$, $p = .002$, $\eta^2 = .06$; conservatism: Pillai’s trace = .099, $F(2, 200) = 10.96$, $p < .001$, $\eta^2 = .10$ (see Table 2).

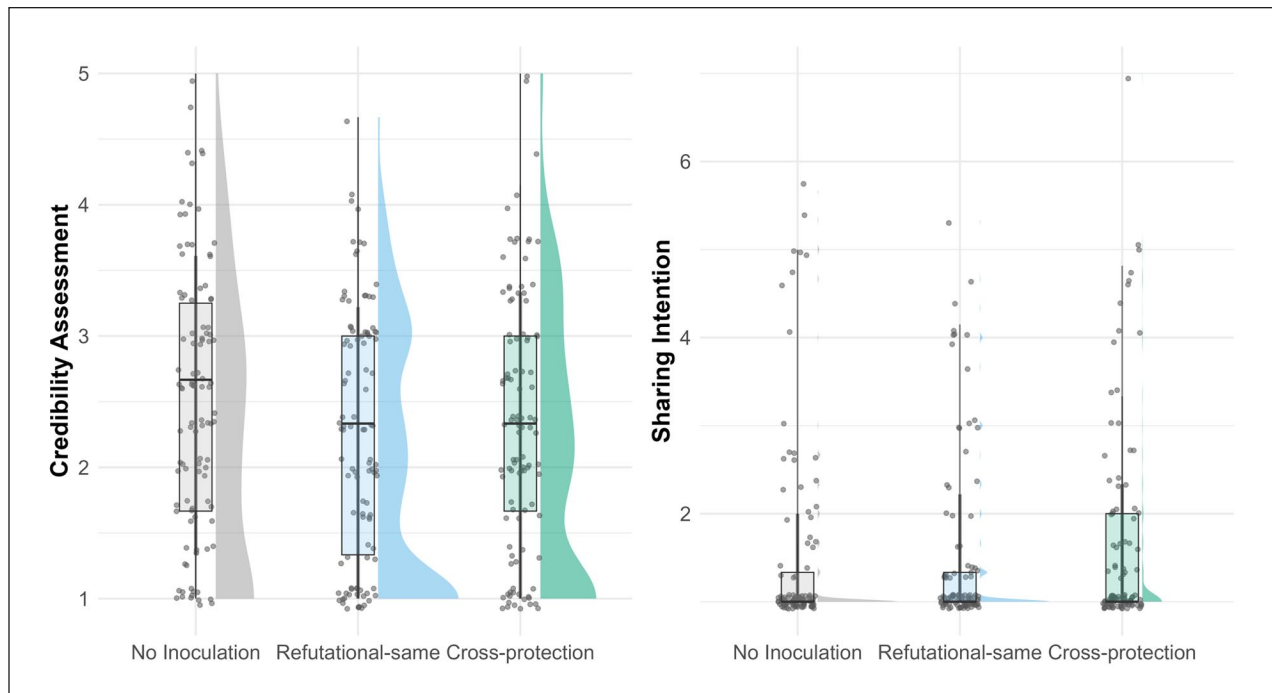


Figure 1. Credibility assessment and sharing intention of misinformation by inoculation condition.

Note. This figure presents the distribution of credibility assessment and sharing intention by experimental manipulation (no inoculation vs. refutational-same inoculation vs. cross-protection condition).

Table 2. Effects of Refutational-Same Inoculation (vs. No Inoculation) on Credibility and Sharing Intention.

Predictor	Pillai's trace	$F(2,200)$	p	η_p^2
Inoculation (same vs. control)	.030	3.10	.047	.03
Progressivism	.061	6.53	.002	.06
Conservatism	.099	10.96	<.001	.10

Note. Multivariate tests are based on Pillai's Trace. Degrees of freedom for each F -test are in parentheses. η_p^2 = partial eta squared (effect size). Model controls for progressivism and conservatism.

Table 3. Univariate Follow-Up: Effects of Refutational-same Inoculation (vs. No Inoculation) on Credibility.

Predictor	$F(2,201)$	p	η_p^2
Inoculation (same vs. control)	4.65	.032	.02
Progressivism	11.58	<.001	.05
Conservatism	18.47	<.001	.08

Note. Univariate follow-up test for credibility. The model controls for progressivism and conservatism. η_p^2 = partial eta squared.

Univariate follow-up analyses showed that the effect of inoculation was driven by credibility, $F(1, 201)=4.65$, $p=.032$, $\eta^2=.02$, whereas no effect emerged for sharing intention, $F(1, 201)<0.01$, $p=.953$, $\eta^2<.001$. Thus, participants who received a refutational-same inoculation rated

misinformation as less credible than participants in the control condition (see Table 3). Accordingly, H1 was supported for perceived credibility, whereas participants' intention to share misinformation was unaffected (RQ1).

We further examined the topic of misinformation (Ukrainian refugees and climate change) separately. For Ukrainian refugees, the multivariate effect of inoculation was significant, Pillai's trace = .073, $F(2, 96)=3.79$, $p=.026$; univariate tests showed a credibility-reducing effect, $F(1, 97)=7.30$, $p=.008$, $\eta^2=.07$, but no effect on sharing intention, $F(1, 97)=0.69$, $p=.408$, $\eta^2=.01$. For climate change, the multivariate effect was not significant, Pillai's trace = .022, $F(2, 99)=1.10$, $p=.338$, $\eta^2=.01$. Taken together this suggests that the observed credibility benefit of the refutational-same inoculation was driven by the Ukrainian refugee misinformation.

H2: Cross-Protection Condition. For cross-protection versus control, the multivariate effect was marginal, Pillai's trace = .028, $F(2, 200)=2.89$, $p=.058$, $\eta^2<.001$, and univariate tests were non-significant for perceived credibility, $F(1, 201)=1.81$, $p=.181$, $\eta^2<.00$, and sharing intention, $F(1, 201)=0.95$, $p=.331$, $\eta^2<.001$. Thus, the cross-protection condition did not significantly reduce either the perceived credibility of misinformation or participants' willingness to share it.

Control Variable: Political Orientation. Political orientation was a significant multivariate predictor in both models (progressivism and conservatism: all $ps<.01$) and univariately for

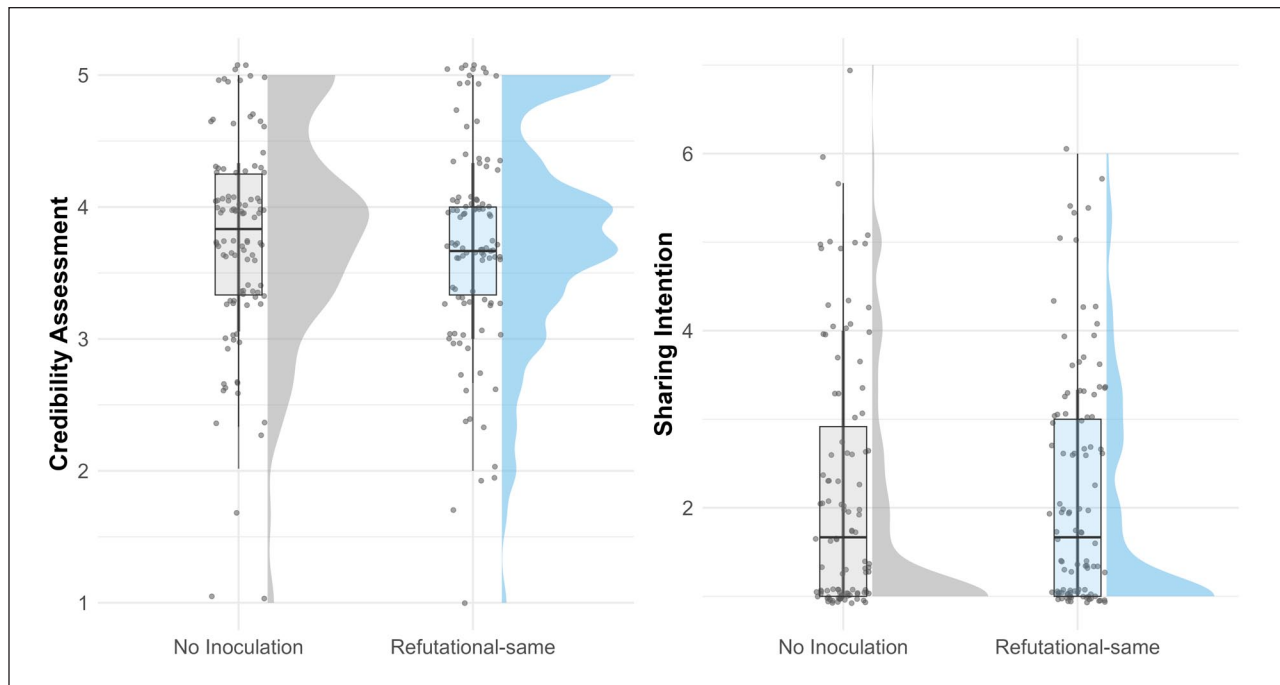


Figure 2. Credibility assessment and sharing intention of true information after no inoculation versus refutational-same inoculation.
 Note. This figure presents the distribution of credibility assessment and sharing intention after no inoculation versus refutational-same inoculation.

both outcomes (all $ps < .01$). Directionally, higher conservatism consistently predicted higher credibility assessments and greater sharing intentions, whereas progressivism was associated with lower perceived credibility and lower sharing intention for misinformation, although its effects were less consistent across models.

RQ1 and H3: Effects of Refutational-Same Inoculation on True Information

An MANOVA tested whether a refutational-same inoculation, presented before a true message, affected perceived credibility (RQ1) or sharing intention (H3) relative to control, controlling for political orientation. The multivariate effect of inoculation was not significant, Pillai's trace = .001, $F(2, 200) = 0.08$, $p = .923$. Univariate tests likewise showed no effect on credibility, $F(1, 201) = 0.10$, $p = .753$, $\eta^2 < .00$, and no effect on sharing intention, $F(1, 201) = 0.02$, $p = .888$, $\eta^2 < .00$. Thus, the refutational-same inoculation did not affect how participants evaluated or intended to share true information, supporting H3 and suggesting no effect for RQ1.

Political orientation showed systematic associations: progressivism was a significant multivariate predictor, Pillai's trace = .110, $F(2, 200) = 12.40$, $p < .001$, and univariately predicted higher perceived credibility of true information, $F(1, 201) = 24.64$, $p < .001$, $\eta^2 = .11$. Conservatism showed no significant multivariate effect, Pillai's trace = .019, $F(2, 200) = 1.94$, $p = .146$, $\eta^2 < .00$, and univariately showed no effect on credibility, $F(1, 201) = 0.32$, $p = .573$, $\eta^2 < .00$, but

was a small positive predictor of sharing intention, $F(1, 201) = 3.90$, $p = .050$, $\eta^2 = .02$.

Overall, these results indicate that refutational inoculations did not reduce the credibility or sharing of true information (see Figure 2), whereas political orientation showed selective associations with the evaluation of true messages.

H4–H7: Moderation Analyses

To test H4–H7, we ran separate moderation models in PROCESS for R (Model 1; Hayes, 2024), with credibility as the outcome, inoculation (refutational-same vs. control) as the predictor, and one moderator per model: CRT, CTDS, NFC, IL, or use of messenger services. Political orientation (progressivism, conservatism) was entered as covariates. All continuous variables were mean-centered. Across models, none of the highest-order interaction reached significance, indicating no evidence that the inoculation effect varied by these individual differences (all $\Delta R^2 \leq .007$).

The interaction for both CRT ($b = -0.024$, $SE = 0.0665$, $t = -0.38$, $p = .708$, $\Delta R^2 = .001$) and CTDS ($b = 0.021$, $SE = 0.031$, $t = 0.66$, $p = .511$, $\Delta R^2 = .002$) was non-significant. Small negative main-effect trends emerged for CRT ($p = .153$) and CTDS ($p = .069$). Thus, the data do not support H4.

The interaction for NFC ($b = -0.092$, $SE = 0.189$, $t = -0.49$, $p = .628$, $\Delta R^2 = .001$) and main effect ($p = .072$) were non-significant, hence, not supporting H5.

The interaction for IL was non-significant ($b = -0.052$, $SE = 0.138$, $t = -0.38$, $p = .707$, $\Delta R^2 = .001$), although higher

Table 4. Interactions Between Inoculation and Individual Differences Predicting Perceived Credibility.

Moderator	B	SE	t	p	ΔR^2
Critical thinking					
Cognitive reflection (H4)	−0.024	0.065	−0.38	.708	.001
Critical thinking disposition (H4)	0.021	0.031	0.66	.511	.002
Need for cognition (H5)	−0.092	0.189	−0.49	.628	.001
Information literacy (H6)	−0.052	0.138	−0.38	.707	.001
Use of messenger services (H7)	−0.129	0.101	−1.27	.204	.007

Note. b = unstandardized coefficient from moderation models (PROCESS for R v4.3.1, Model 1) with credibility as the outcome and inoculation (refutational-same vs. control) as the predictor. Progressivism and conservatism were included as covariates. All continuous predictors were mean-centered. ΔR^2 refers to the change in explained variance due to the interaction term.

IL predicted lower credibility overall ($b = -0.178$, $SE = 0.071$, $t = -2.52$, $p = .013$). Nevertheless, H6 was not supported.

The interaction for use of messenger services was non-significant ($b = -0.129$, $SE = 0.101$, $t = -1.27$, $p = .209$, $\Delta R^2 = .007$), although messenger use showed a marginally positive main effect ($p = .097$). Therefore, not supporting H7. For an overview of all results, see Table 4.

Across all moderation models, the refutational-same inoculation main effect was consistently negative and reached conventional significance in the PROCESS models ($ps \leq .018$), while conservatism positively predicted perceived credibility of misinformation ($ps \leq .001$) and progressivism was not significant. Overall, the data provide no evidence that the inoculation effect depends on any of the tested moderators; the direction of the inoculation effect on credibility was consistent across models.

Discussion

In this online experiment, we investigated whether refutational-same and cross-protection conditions influence the perceived credibility and sharing intentions for false and true information within messenger communication. We further looked at whether these effects are influenced by individual factors such as critical thinking abilities (CRT, CTDS, NFC), IL, reliance on messenger services as news source, and political orientation.

Refutational-Same Inoculation and Perceived Credibility of Misinformation

Consistent with H1, the refutational-same inoculation produced a small but reliable reduction in the perceived credibility of misinformation, while having no influence on sharing intention. Numerous previous studies have shown that inoculation can lead to more resistance to misinformation compared to no inoculation (Lu et al., 2023; Piltch-Loeb et al., 2022). Conceptually, this aligns with classic inoculation logic (McGuire, 1981/1964): pre-exposure plus refutational forewarning equips recipients to discount specific misleading claims (e.g., via perceived threat and accessible

counterarguments; Compton & Pfau, 2005). Practically, the effect appears modest, with topic-level analyses clarifying where it emerges: the credibility reduction was driven by misinformation about Ukrainian refugees, aligning with research on inoculation against Russian disinformation-campaigns (Ziemer et al., 2024). The effect was absent for climate-change misinformation, suggesting topic specificity rather than a generalizable protection effect. Given that climate change is highly polarized (Druckman & McGrath, 2019), strong prior attitudes and underlying political preferences may have constrained inoculation effects. For instance, 21% of the people in West Germany and 25% of the people in East Germany do not believe in anthropogenic climate change (Zandt, 2023). More broadly, these topic-specific patterns may partly reflect the German context, including prevailing attitudes toward climate policy and refugee-related issues and the local media and political landscape; however, the present design does not allow us to disentangle contextual from more general mechanisms.

The cross-protection condition showed no evidence of a generalizing effect for credibility or sharing. This finding does not support the assumption that broader inoculations enhance deeper processing and recognition of misinformation (Cook et al., 2017; Roozenbeek et al., 2022a). It aligns with mixed evidence that broad-spectrum or content-dissimilar inoculations do not reliably transfer to novel topics without sufficient conceptual overlap or guided counterarguing practice (e.g., Parker et al., 2016; Zerback et al., 2021). Moreover, Parker et al., (2012) found that cross-protection effects are stronger when the inoculated issue and the subsequent attack share conceptual similarity, which may explain the null effect here. Taken together, our data indicate that effective protection against misinformation in messenger communication is specific, not generic.

Regarding sharing intention, neither refutational-same nor the cross-protection condition had a protective effect, aligning with prior research (Lu et al., 2023). One interpretation is that our inoculation message primarily operated on belief revision, whereas sharing also reflects social and emotional motives that were not targeted. Sharing motives that are associated with the spread of misinformation are

self-disclosure or fear of missing out (FOMO; Talwar et al., 2019), perceived relevance of information (Apuke & Omar, 2021; Bobkowski, 2015), status-seeking (Kalogeropoulos, 2021), or elicited negative emotions (McLoughlin et al., 2024; Wintterlin et al., 2023). A second, pragmatic consideration is that baseline sharing intention was low, leaving limited room for reduction.

Overall, refutational-same inoculation reduced perceived credibility (particularly for the Ukrainian refugee topic) but did not reduce sharing intention of misinformation. Because MIMS have distinct properties (e.g., end-to-end encryption) that protect private communication, these findings raise questions about the practicality of topic-specific content inspection and underscore the need to explore more privacy-preserving techniques (e.g., broad-spectrum inoculation; Panahi et al., 2025).

No Effects of Refutational-Same Inoculation on True Information

Refutational-same inoculations did not affect either the perceived credibility or the sharing intention of true information. This is a desirable outcome, as measures against misinformation should not inadvertently depress belief in accurate content or induce generalized over-skepticism. The finding supports meta-analytic evidence that inoculation does not undermine credibility judgments of true information (Lu et al., 2023), although it contrasts with reports that some gamified inoculations can prompt more conservative responding (i.e., judging true items as false; Modirrousta-Galian & Higham, 2023). It also aligns with work showing that broad-spectrum inoculation videos can improve discrimination between true and false content (Roozenbeek et al., 2022b). One plausible interpretation is that inoculation increases deeper engagement with information (Cook et al., 2017) and activates counterarguing routines (Compton, 2013) that fail to produce valid counterarguments when the information is accurate, thereby leaving acceptance of true information intact. The absence of any reduction in sharing intention for true information suggests that inoculation does not discourage the dissemination of accurate content, although we also observe no positive effect on sharing. This would be desirable in principle, as inoculation has, in some domains, been associated with increased sharing of legitimate, pro-health information (Park et al., 2022). The null results may be explained by the generally low baseline sharing intention in our sample. Likewise, inoculation may be insufficient to reduce sharing of any kind of message, as it may be driven by motives beyond credibility (Lu et al., 2023; Talwar et al., 2019). The results should, therefore, be interpreted with caution.

Overall, the null effects for true information indicate that the intervention preserved trust in accurate content and, although relatively low, willingness to share it. Addressing

this concern about unintended backfire is important, as countermeasures should target where they are needed most: misinformation.

No Evidence That Refutational-Same Inoculation Effects Depends on Critical Thinking, Information Literacy, or Messenger Use

We tested whether the effect of refutational-same inoculation on credibility of misinformation varied by critical thinking abilities (cognitive reflection, critical thinking disposition, and NFC), IL, or use of messenger services. Results indicate that the credibility-reducing effect of inoculation does not depend on any tested moderator. Below, we discuss each moderator briefly.

We assumed that higher critical thinking abilities would foster deeper engagement with the inoculation messages and thereby increase the inoculation's impact on the perceived credibility of misinformation. Our findings do not support this. Although critical thinking is often linked to better detection of misinformation (Escolà-Gascón et al., 2021; Orhan, 2023) via reflective elaboration (McPeck, 1981) and analytic processing (Amer, 2005; Evans & Stanovich, 2013; Pennycook & Rand, 2019), neither cognitive reflection nor critical thinking disposition moderated the effect of refutational-same inoculations, aligning with previous findings (Roozenbeek et al., 2022b). Research also shows that higher NFC fosters deeper processing and source scrutiny (Cacioppo et al., 1996; Mokhtari et al., 2013) and enhances misinformation recognition (Schaewitz et al., 2020). Contrary to expectations, NFC did not moderate the effect of refutational-same inoculations, which may reflect lower task involvement or inoculation messages' simplicity that reduced variance in cognitive effort (Pfau et al., 1997). The absence of moderation by critical thinking may also be explained by the inoculation itself: simply presenting an inoculation can supply ready-made counterarguments, reducing the need to engage in effortful critical thinking. At the same time, processing of both the inoculation and the subsequent message may be guided by other motives, most notably political orientation. In our models, conservatism consistently predicted higher credibility ratings of misinformation, whereas progressivism did not (in these models), indicating an asymmetry in how political leanings relate to credibility assessments. This pattern is consistent with motivated reasoning (Kunda, 1990), with effects that may depend on cognitive abilities and the specific content presented (Strömbäck et al., 2024).

Despite benefits for identifying credible sources and empirical links to reduced misinformation belief (Al Zou'bi, 2022; Jones-Jang et al., 2021), IL did not moderate the effect of refutational-same inoculation, although higher IL related to lower credibility overall. This may be explained by self-report measurement limiting sensitivity.

Although heavier social media and messenger use can promote belief in conspiracy theories and misinformation (Allington et al., 2021; Enders et al., 2023; Theocharis et al., 2021), messenger use did not moderate inoculation, although a marginally positive main effect suggests slightly higher credibility among heavier users. This might be explained by using a single-item measure, as multiple-item measures have better reliability and criterion validity (Sarstedt & Wilczynski, 2009). Moreover, we did not differentiate between different messenger services (e.g., Facebook Messenger vs. WhatsApp). Previous findings suggest that the use of Facebook Messenger is positively related to belief in misinformation, whereas no significant relationship was found between WhatsApp use and belief in misinformation (Altay et al., 2024).

Overall, there was no evidence that the (negative) effect of refutational-same inoculation on perceived credibility varied across the examined cognitive and media-use differences, suggesting that it does not depend on recipients' critical thinking abilities, IL, or messenger use.

The Influence of Political Orientation

Across analyses, and consistent with prior work linking political ideology to belief in misinformation (Baptista et al., 2021; Calvillo et al., 2020), political orientation consistently predicted credibility assessments and sharing intentions. For misinformation, higher conservatism predicted higher credibility and greater sharing intention. This pattern aligns with motivated reasoning (Kunda, 1990), in which information processing can be shaped by accuracy motives (favoring accurate conclusions) or directional motives (favoring preference-consistent conclusions). Empirically, preference-inconsistent information is more likely to be rejected as false (Tsang, 2021), whereas preference-consistent information is accepted more readily (Kahne & Bowyer, 2017). Moreover, corrections of misinformation seem to be more effective when preference-consistent (Hart & Nisbet, 2012; Nyhan & Reifler, 2010). In MIMS, repeated exposure to preference-consistent content can increasingly misinform partisan users (Rossini & Kalogeropoulos, 2025). Taken together, these patterns may contextualize why inoculation effects can appear topic-specific. When they clash with recipients' priors (e.g., on climate change), they may invite effortful counterarguing; when they align, they may encounter less scrutiny. However, we did not test interactions between inoculation and political orientation, so this interpretation is speculative. Designing inoculation messages that activate accuracy motives while minimizing identity threat may be a promising direction.

Limitations

This study has several limitations. First, although the experimental conditions were operationalized in line with the two-component structure of inoculation theory, perceived threat

was not assessed directly. Consequently, we cannot confirm whether threat was activated as intended or attribute the observed effects uniquely to inoculation rather than adjacent processes such as skepticism or critical thinking. Moreover, the forewarning was formulated in a relatively general manner to reflect the MIMS context and the multi-topic design. This may have elicited weaker perceptions of threat compared to more explicit or issue-specific forewarnings used in some inoculation studies. Second, most measures relied on self-reports, which are vulnerable to bias (e.g., social desirability). Third, we measured sharing intention rather than observed sharing behavior; whether participants would actually share information in more realistic settings remains uncertain. Fourth, the messenger chat simulation was deliberately constrained (screenshot-based and non-interactive): participants could not react to or share messages or observe others' reactions, and sender cues (e.g., expertise, relational tie) were masked. These design choices increase internal control but reduce ecological validity for MIMS dynamics. Moreover, the intervention design of inoculation messages may have lacked salience or perceived credibility (e.g., no source attribution), potentially attenuating effects. Alternative message designs and integration within chats could yield different outcomes. An additional limitation concerns statistical power. Although the sample size approximated the a priori power analysis for detecting the anticipated main effects, interaction effects generally require larger samples to achieve adequate power. Accordingly, null findings should be interpreted with caution, as the study may have been underpowered to detect smaller (interaction) effects. Moreover, our German-speaking sample limits cultural generalizability; attitudes toward the misinformation topics (e.g., Ukrainian refugees, climate change) as well as media use may vary across countries. Finally, because baseline attitudes toward the specific topics were not measured prior to exposure, some participants may have encountered the messages in a manner closer to therapeutic rather than prophylactic inoculation, which future research could address by measuring initial attitudes more explicitly.

Future Research

Our findings indicate that refutational-same inoculation can reduce credibility assessments of misinformation, yet its applicability in MIMS is constrained: private messaging is and should remain off limits to surveillance given the fundamental right to confidential correspondence. Accordingly, future work should test privacy-protective approaches, such as broad-spectrum inoculation that builds generalizable skills (e.g., media literacy and recognition of manipulation techniques) and compare them with narrow, topic-specific inoculations in MIMS-like settings. Such research should consider empirical findings, technical considerations, and fundamental rights equally, while not solely placing

responsibility on individuals but also on service providers and policymakers (see Panahi et al., 2025). Methodologically, studies should explicitly test whether political orientation moderates inoculation effectiveness, employ more realistic environments (e.g., interactive chat simulations) that capture actual sharing behavior, and use performance-based measures alongside self-reports to improve sensitivity.

Conclusion

This study examined whether inoculation-based measures can reduce belief in and sharing of misinformation in messenger communication. Consistent with prior inoculation research, refutational-same inoculation reduced the perceived credibility of misinformation but not sharing intention. Topic-level analyses showed that this effect was evident for misinformation about Ukrainian refugees and absent for climate change, pointing to topic specificity rather than broad generalizability.

The cross-protection condition did not influence credibility assessments or sharing intention, suggesting no evidence of a generalized protective effect. For true information, refutational-same inoculation left both credibility and sharing intention unchanged, indicating that the intervention did not promote generalized skepticism toward accurate content. No moderation effects emerged for critical thinking abilities, IL, or messenger-based news use, whereas political orientation consistently predicted perceived credibility and sharing intentions.

Taken together, these findings suggest that refutational-same inoculation in MIMS may help reduce the perceived credibility of misinformation under some conditions, while offering limited evidence for broader or behavioral effects. In private messaging environments, where communication is and should remain protected, these results raise practical and ethical questions about implementation. Future research should therefore assess whether, and in what form, inoculation-based interventions are feasible and worthwhile in MIMS, with careful attention to privacy, topic specificity, and real-world sharing behavior.

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Ethical Considerations

This study was approved by the University of Duisburg-Essen Research Ethics Committee (ID 2402SPZJ2881) on February 20, 2024.

Consent to Participate

Participants gave written consent before starting the experiment.

Consent for Publication

Not applicable.

Author Contributions

Amancay Ancina: Project administration; Conceptualization; Methodology; Validation; Investigation; Formal analysis; Writing – Original Draft; Writing – Review & Editing; Visualization; Supervision.

Jana Zumbärgel: Investigation; Conceptualization; Methodology; Resources; Writing – Original Draft.

Nicole Krämer: Conceptualization; Methodology; Writing – Review & Editing; Supervision; Funding acquisition.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The experiment and hypotheses were preregistered (<https://osf.io/yfhn7/overview>); all measures, stimulus material, and data are available on the Open Science Framework (OSF; <https://osf.io/pevuy/overview>).

Supplemental Material

Supplemental material for this article is available online.

Notes

1. <https://osf.io/yfhn7/overview>
2. <https://osf.io/pevuy/overview>
3. BAföG (*Bundesausbildungsförderungsgesetz*) is Germany's federal student aid program.

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